PYTHON PRACTICE PROGRAMS

Strings • Lists • Tuples • Sets • Dictionaries

1. Reverse a string Consonants
Question: Accept a string from the user and display it in reverse order.

2. Count Vowels and Consonants
Question: Count the vowels and consonants in a string. Ignore non-alphabetic characters.

3. Check Whether a String Is a Palindrome
Question: Check whether a word or phrase is a palindrome, ignoring spaces and letter case.

4. Count Words, Characters, and Spaces
Question: Accept a sentence and count its words, characters, and spaces.

5. Frequency of Each Character
Question: Count how many times each character occurs in a string.

6. Basic List Operations
Question: Accept five integers in a list and display their sum, average, maximum, and minimum.

7. Remove Duplicates While Preserving Order
Question: Remove duplicate elements from a list while keeping the original order.

8. Sort a List
Question: Accept numbers from the user and display them in ascending and descending order.

9. Separate Even and Odd Numbers
Question: Separate the even and odd numbers from a list into two lists.

10. Demonstrate List Methods
Question: Demonstrate append(), insert(), remove(), pop(), sort(), and reverse().

11. Sum, Maximum, Minimum, and Length of a Tuple
Question: Create a tuple of numbers and calculate its sum, maximum, minimum, and length.

12. Count Occurrences in a Tuple
Question: Count how many times an element appears in a tuple and find its first index.

13. Convert a List into a Tuple and Back
Question: Convert a list into a tuple, then convert the tuple back into a list.

14. Perform Set Operations
Question: Perform union, intersection, difference, and symmetric difference on two sets.

15. Remove Duplicates Using a Set
Question: Accept a list of numbers and remove duplicate elements using a set.

16. Check Subset and Super set
Question: Check whether one set is a subset or superset of another set.

17. Student Marks Dictionary
Question: Store five students' names and marks in a dictionary, display them, and calculate average marks.

18. Find the Student with the Highest Marks
Question: Identify the student who has obtained the highest marks from a dictionary.

19. Demonstrate Dictionary Methods
Question: Demonstrate keys(), values(), items(), get(), update(), and pop().

20. Count Word Frequency
Question: Count the frequency of each word in a sentence, treating uppercase and lowercase words as the same.

Quick Revision: Topics Covered

Topic	Programs	Key Concepts
Strings	1–5	Slicing, loops, split(), count(), character frequency
Lists	6–10	Input, append(), insert(), remove(), sorting
Tuples	11–13	count(), index(), tuple(), list()
Sets	14–16	Union, intersection, difference, subset, superset
Dictionaries	17–20	Keys, values, updating, searching, frequency counting

Suggested Lab Activity: Write each program independently, execute it with at least three different test cases where applicable, and explain the output and the purpose of each method used.